MIL-H-80009B
26 February 1974
SUPERSEDING
MIL-H-80009A
8 July 1968 and
MIL-H-14253C(WC)
11 February 1970

MILITARY SPECIFICATION

*

HONING MACHINES, HORIZONTAL, SINGLE SPINDLE, MANDREL TYPE, WET APPLICATION

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

* 1.1 Scope. This specification covers electric motor driven, bench and floor mounted honing machines with equipment and accessories as described herein.

* 1.2 Classification. Honing machines covered by this specification are of the following types, styles, and sizes. The machine supplied shall be of the type, style, and size specified (see 6.2.1).

Type I - Bench Mounted

Size	Bore Diameter, Inches, Maximum	Bore Length, Inches
312	3	12

Type 11 - Floor Mounted

Style A - Hand Held - Manually Stroked

Size	Bore Diameter, Inches, Maximum	Bore Length, Inches
312	3	12
618	6	18

Style B - Workholder Held - Power Stroked

Size	Bore Diameter, Inches, Maximum	Bore Length, Inches
318	3	18

FSC 3419

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2. APPLICABLE DOCUMENTS

2.1 Government publications. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extend specified herein.

FEDERAL

STANDARD

FED-STD-595	Colors

MILITARY

SPECIFICATIONS

MIL-M-18058	Machinery, Metal and Woodworking, Support Equipment and Associated Repair Parts, Preparation for Delivery of

STANDARDS

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U. S. Military Property
MIL-STD-461	Electromagnetic Interference Characteristics Requirements for Equipment

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

* 2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

OCCUPATIONAL SAFETY AND HEALTH ACT (OSHA) OF 1970

Title 29, Code of Federal Regulations, Chapter XVII, Part 1910, and amendments.	Occupational Safety and Health Administration Standards

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NATIONAL BUREAU OF STANDARDS (NBS) HANDBOOK H28

Screw-Thread Standards for Federal Services

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

JIC ELECTRICAL STANDARDS FOR GENERAL PURPOSE MACHINE TOOLS

(Application for copies should be addressed to the National Machine Tool Builders' Association, 7901 Westpark Drive, McLean, VA 22101.)

AMERICAN GEAR MANUFACTURERS ASSOCIATION AGMA 360

Manual for Machine Tool Gearing

(Application for copies should be addressed to the American Gear Manufacturers Association, 1330 Massachusetts Ave., NW, Washington, DC 20005.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal Agencies.)

3. REQUIREMENTS

* 3.1 First Article approval. If First Article approval is required (see 6.2.1 and 6.3), inspection therefor shall be in accordance with Section 4. A First Article comprises a preproduction model or initial production sample; approval involves testing and evaluating the First Article for conformance with the requirements herein before, or in the initial stage of, production under a contract.

3.2 Design. The machine shall be new and one of the manufacturer's current models . The machine shall be capable of honing holes in accordance with the requirements herein. The design and construction shall include all components, parts, and features necessary to meet the performance requirements specified herein. All parts subject to wear, breakage, or distortion shall be accessible for adjustment, replacement and repair. Type I and Type II, Style A machines shall be designed for hand held, manually stroked workpieces, and shall have means for absorbing the rotational torque transmitted to the workpiece. Type II, Style B machines shall have a workholding device and a power stroking mechanism that will automatically stroke the workpiece without operator attention, and shall also be operable in the manually stroked mode by disengaging the automatic control and moving the workholder and power stroking mechanism out of the normal working position.

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3.2.1 Controls. All operating controls shall be grouped in a location convenient to the operator except as otherwise specified herein.

* 3.2.2 Safety and health requirements. Covers or other safety devices shall be provided for all parts of the machine presenting safety hazards which can be guarded without substantial interference to operation of the machine. All guards shall provide easy access to guarded parts. Except as otherwise specified by the procuring activity, the machine shall comply with the standards promulgated under OSHA applicable to the machine manufacturer. Exceptions and additional requirements for safety and health shall be as specified (see 6.2.1 and 6.5).

3.2.3 Lubrication. Means shall be provided to insure adequate lubrication for all moving parts. Recirculating systems shall include filters which are cleanable. Each lubricant reservoir shall have means for determining fluid level. All oil holes , grease fittings and filler caps shall be accessible.

3.2.4 Interchangeability. To provide for replacement of worn parts, all parts bearing the same part number shall be functionally interchangeable, and shall be dimensionally identical within the manufacturing tolerance limits in use by their manufacturer.

3.3 Construction. The machine shall be so constructed that when installed and connected to power, it can perform to its maximum functional capacity. The machine shall be constructed of parts which are without any defect and free of repair. The structure shall be capable of withstanding all forces encountered during operation of the machine to its maximum rating and capacity without permanent distortion.

3.3.1 Castings and forgings. All castings and forgings shall be free of defects, scale, and mismatching. No process such as welding, peening, plugging, or filling with solder or paste shall be used for reclaiming any defective part.

3.3.2 Welding, brazing, or soldering. Welding, brazing, or soldering shall be employed only where specified in the original design. None of these operations shall be employed as a repair measure for any defective part.

3.3.3 Fastening devices. All screws, pins, bolts, and similar parts shall be installed in such a manner as to prevent change of tightness. Those subject to removal or adjustment shall not be swaged, peened, staked, or otherwise permanently deformed.

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3.3.4 Surfaces. All surfaces of castings, forgings, molded parts, stampings, and welded parts shall be cleaned and free from sand, dirt, fins, sprues, flash, scale, flux and other harmful or extraneous materials. All edges shall be either rounded or beveled unless sharpness is required to perform a necessary function. Except as otherwise specified herein, the condition and finish of all surfaces shall be commensurate with the manufacturer's standard commercial practice.

3.3.5 Painting. Unless otherwise specified (see 6.2.1), the machine shall be painted in accordance with the manufacturer's standard commercial practice. When specified, the final color shall be Standard Machine Tool Light Gray similar to No. 16187 Gray Gloss, FED-STD-595 or NMTBA No. 7-B.

3.3.6 Threads. All threaded parts shall be in the inch system and shall conform to Handbook H28.

3.3.7 Plates. All words on speed and feed indicating plates and on instruction plates shall be in the English language, engraved, etched, embossed, or stamped in boldface letters on a contrasting background.

* 3.3.8 Gears. All gears used in the machine and its components shall be machined in the inch system and shall conform to the applicable requirements of AGMA 360.

* 3.4 Components. Each machine shall consist primarily of a spindle, spindle drive mechanism, stone expansion mechanism, cutting pressure control, sizing control, honing oil system, and an enclosure to house the above components. Type I machines shall have means for securing to a suitable workbench. Type 11 machines shall have a base with provisions for securing to the floor.

* 3.4.1 Spindle. The spindle shall be mounted on ball or roller bearings which are sealed to prevent entrance of dirt, coolant, or abrasives. The spindle shall have means for holding and driving all sizes of mandrels necessary to hone holes within the range specified in Table I. The spindle and bearings shall be manufactured to the accuracy necessary to produce cylindrical holes within the accuracies required herein. Type II, Style B machines shall have in addition to the provisions of 3.4, self-contained means for stroking the part relative to the honing tool, and automatically terminating the honing operation at the end of the honing cycle.

3.4.2 Spindle drive mechanism. The spindle shall be driven by an electric motor. The drive mechanism shall consist of either a belt drive system, or an infinitely variable speed control mechanism. Belt drive systems shall have either an automatic or a manual belt tension adjustment device. Both types of drive systems shall have means for determining spindle speed.

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* 3.4.3 Stone expansion mechanism. Each machine shall be equipped with a stone expansion system. Type 1 machines shall have a foot control that starts and stops spindle rotation and retracts the stone, when disengaged, to facilitate removal of the workpiece from the honing mandrel. The foot pedal may also control the stone expansion rate and cutting pressure, by varied force exerted on the foot pedal, if cutting pressure is not otherwise provided by the machine. Type II machines shall incorporate either a hydraulic or mechanical stone expansion system, as specified (see 6.2.1). The hydraulic system shall operate in conjunction with a foot control, but the expansion rate shall be controlled by a dial which adjusts the hydraulic pressure to maintain desired stone pressure. The mechanical system shall operate a cutting pressure control that will automatically expand the stones to maintain a preselected stone pressure. The system shall operate in conjunction with a foot control, but the stone expansion rate shall be independent of the amount of force exerted on the foot control On Type II, Style B machines, when operating in the manual stroke mode, and on Type II, Style A machines, the foot control shall start and stop spindle rotation, and when released, shall retract the honing stone to facilitate workpiece removal from the mandrel. On Type II, Style B machines, when operating in the automatic mode, depressing the foot pedal shall start spindle rotation and shall initiate stone expansion and power stroking; the automatic controls shall automatically stop spindle rotation and retract the honing stone after the bore diameter has been honed to proper size without requiring further manipulation of the foot control.

* 3.4.4 Cutting pressure control system. Each honing machine shall have infinitely variable cutting pressure within the limits of the machine, and shall provide a range of cutting pressures suitable for finish honing of small parts to those pressures required for rough honing of larger parts. Hydraulically operated stone pressure controls shall provide a dial for adjusting stone pressure and a gage for monitoring the pressure. Mechanically operated systems shall provide a spring loaded pressure control having a calibrated dial for adjusting the stone pressure.

* 3.4.5 Sizing control. Each machine shall be provided with a sizing control. Type I machines shall have a sizing control mechanism that will prevent further stone expansion when the preselected diameter of the workpiece is approached. The mechanism may be actuated by the stone expansion mechanism, in lieu of automatically gaging the workpiece bore, and need not have automatic compensation for stone wear. Type II machines shall have a sizing control of a type that can be preadjusted in accordance with the amount of stock to be removed from the workpiece. A dial shall indicate continuously how much stock is yet to be removed as the bore approaches the desired size, to apprise the operator when a more accurate gaging device is necessary. The indicating dial may be activated by the stone expansion mechanism, and need not have automatic stone wear compensation In addition,

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at the option of the operator, an automatic device may be activated which shall lock the machine in honing position and automatically release the control to stop the honing action when the bore approaches the desired size and when more accurate gaging device is necessary. The operator shall initiate the hone cycle, and retain the option to override the automatic control at any time.

* 3.4.6 Honing oil system. Each machine shall have a honing oil system consisting of a reservoir, a filter or sludge removal system, a pump, oil delivery nozzle or petcock, and a pan for collecting spent coolant and returning it to the reservoir. The pump shall be driven by a separate motor, or the main drive motor, as indicated in Table I. The pump discharge capacity shall be adequate for honing parts within the specified size range of the machine. The oil delivery nozzle(s) shall have petcocks or other means for controlling the volume of oil and for directing the flow into the workpiece. The splash pan and guards shall be designed to minimize splash. The filter or sludge removal system shall be located between the coolant pan and the pump inlet and shall remove abrasive and metallic particles which might damage the pump.

3.4.6.1 Pressurized honing oil filtering system When specified (see 6.2.1), the machine shall have a pressurized honing oil filtering system meeting the following requirements. The pressurized honing oil system shall have a reservoir conforming to the requirements of Table I and a self contained means for pressurized filtering and regulating the flow of honing oil. It shall have a suction strainer capable of removing all abrasive and metallic particles of a size which might damage the pump. The pump shall develop adequate pressure to force the honing oil through a replaceable, cartridge type filter and shall be provided with a relief valve set to a pressure that will prevent rupture of a cartridge when plugged. The filter system shall be capable of delivering 1½ GPM at 20 psi back pressure, shall have not less than 4000 square inches of filter surface, and shall have a sludge capacity not less than 70 cubic inches. The filter shall be capable of filtering the honing oil to the degree necessary to enable the machine to hone to a 5 microinch finish.

* 3.4.7 Work support. Each Type I and Type II, Style A machine shall be equipped with work support bars to absorb torque transmitted from the hone to the workpiece. The work support bar shall be not less than one inch longer than the longest standard mandrel for the machine. The bar height and distance from the mandrel shall be adjustable to accommodate various workpieces. Each Type II, Style B machine shall have a universal work support that is moved by power in a stroking motion relative to the spindle. Power shall be provided by a separate motor. The work support shall be of the universal type, adaptable to a wide variety of workpiece configurations, through the use of commercially available fixturing.

* 3.4.8 Stroking stop . Each Type I and Type II, Style A machine shall be provided with an adjustable stroking stop to prevent stroking the workpiece past the honing stones and to facilitate honing of blind holes and

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stepped bores. Type II, Style B machines are not required to have stroking stops. The stroking mechanism, incorporated in the design of the Type II, Style B machine, shall provide a mechanically or hydraulically controlled stroke length infinitely variable from 1/4″ to 6“ in length, with provisions for varying the position of the stroking pattern over a 4-inch range along the axis of the mandrel.

3.5 Sizes and capacities. The sizes and capacities of the machine shall meet the requirements of Table 1.

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TABLE I - SIZES AND CAPACITIES

CHARACTERISTIC	** REQUIREMENT			
	Type I	Type II Style A		Type II Style B
	Size 312	Size 312	Size 618	Size 318
Bore range, diameter, inches:				
Manual	0.480 to 3.000	0.060 to 3.000	0.060 to 6.000	0.060 to 6.000
Power	-	-	-	0.0125 to 3.875
Bore range, length, inches	1/4 to 12	1/4 to 12	1/4 to 18	1/4 to 18
Spindle speeds, number	4 or infinite	10 or infinite	10	18
Spindle speed range, RPM	450 to 740	250 to 1150	200 to 1600	200 to 2500
Spindle drive horsepower	1/3	1/2	1/2	1
Honing oil pump horsepower	1/15	1/15	From main drive # or 1/15	From main drive or 1/2
Stroking motor horsepower	-	-	-	1/2
Honing oil reservoir, capacity, gallons	3	10	12	24

** Unless otherwise specified (see 6.2.1).
As specified (see 6.2.1)

Requirements are not less than those shown. Where a range is shown, the required performance is a range from the smaller figure or less to the larger figure or more.

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3.6 Accuracy and performance. Each Type I machine and associated tooling shall be capable of honing cylindrical surfaces to an accuracy of 0.0002 inch (TIR) for roundness and within 0.0002 inch constant diameter throughout the length of the bore when tested in accordance with 4.5.2. The machine shall also be capable of producing a surface finish not rougher than 10 microinches when honing a gray iron casting. Each Type II machine and its size control mechanism-shall be capable of producing two duplicate bore sizes to the following tolerances when tested in accordance with 4.5.3.

a. Diameter of finished bores - 1.0050 ± 0.0001 inch.

b. Roundness - The bores shall be not more than 0.00005 inch out of round.

c. Constant diameter - Diameters of finished bore shall not vary more than 0.0001 inch (TIR) throughout 3 inch length of samples.

d. Surface finish - The surface finish shall be not rougher than 5 microinches.

3.7 Electrical system. Unless otherwise specified (see 6.2.1), the electrical system shall be in accordance with this paragraph and the applicable subparagraph. Each machine shall have an identified ground terminal for grounding the machine when installed. The terminal shall be suitable for connection of the size grounding conductor specified in Table 15-1 of JIC Electrical Standards for General Purpose Machine Tools. Each machine system shall include undervoltage protection and a fused safety disconnect switch in accordance with JIC Standards.

3.7.1 Type I. In addition to the requirements in 3.7, each Type I machine shall have an electrical system in accordance with the manufacturer's standard commercial practice. Input power shall be 115 volt single phase 60 Hz.

3.7.2 Type II. In addition to the requirements in 3.7, each Type II machine shall be in accordance with JIC Standards. Input power shall be 230/460 volt 3 phase 60 Hz. The system shall be initially wired for operation on 230 volts.

3.7.3 Motors. Motors shall have ball or roller bearings. Continuous duty horsepower ratings shall conform to the requirements shown in Table I. Unless otherwise specified (see 6.2.1), each motor shall be single speed and shall meet the requirements of a dripproof enclosure.

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3.7.4 Elapsed time indicator. When specified (see 6.2.1), the machine shall have an elapsed time indicator, which shall be a digital readout, non-reset type designed to resist the effects of vibration, voltage fluctuation, temperature and humidity. It shall be mounted on the machine and connected to record the accumulated operating time of the spindle drive motor to a total of not less than 9999,9 hours.

3.8 Hydraulic system. When a hydraulic system is furnished, it shall be complete including all pumps, valves, piping, cylinders, and pressure controls. Overpressure protection shall be provided on the high pressure line from the hydraulic pump to prevent damage to components, A filter system shall be provided to insure delivery of clean fluid to the hydraulic system. The hydraulic reservoir shall have a sight gage to indicate the fluid level.

* 3.9 Standard equipment. Unless otherwise specified (see 6.2.1), the following equipment shall be furnished with each machine.

a. One set of hand tools normally furnished by the manufacturer.

b. Supply of honing oil sufficient for the initial filling of machine.

3.10 Optional equipment. Optional equipment and accessories shall be furnished as specified (see 6.2.1).

3.10.1 Optional tooling. Honing mandrels, honing stones, truing sleeves, and other tooling shall be furnished as specified and fully described (see 6.2.1).

* 3.11 Electromagtic interference control. When specified (see 6.2.1), equipment procured under this specification shall comply with MIL-STD-461 for Class IIB equipment.

3.12 Fungus control. When required (see 6.2.1), fungus proofing shall be as specified.

3.13 Lubrication chart or plate. Unless otherwise specified (see 6.2.1), a lubrication chart or plate shall be permanently and securely attached to each machine. If a chart is furnished, it shall be placed in a transparent plastic folder, or permanently sealed between clear plastic sheets, with suitable means for mounting. The following information shall be furnished on the chart or plate:

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Points of application

Service interval

Type of lubricant

Viscosity

Federal or Military Specification (if available)

* 3.14 Nameplate. Unless otherwise specified (see 6.2.1), a corrosion-resistant metal nameplate identifying the machine in accordance with MIL-ST,D-130 shall be securely attached to each machine. The nameplate shall contain the information listed below. If the machine is a special model, the model designation shall include the model of the basic standard machine and a suffix identified in the manufacturer's permanent records. The captions listed may be shortened or abbreviated, provided the entry for each caption is clear as to its identity.

Nomenclature

Manufacturer's name

Manufacturer's model designation

Manufacturer's serial number

Type_____

Style_____

Size_____

Power input (volts, total amps, phases, frequency)

Contract Number or Order Number

Federal Stock Number or Plant Equipment Code

Date manufactured

u s

* 3.15 Technical data. Data shall be furnished as specified (see 6.2.2).

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3,16 Workmanship. Workmanship of the machine and accessories shall be of a quality equal to that of the manufacturer's commercial equipment of the type specified herein.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or order, the supplier is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the contract order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification when such action is deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 First Article inspection. When First Article approval is required under 3.1, First Article inspection shall be performed. Unless otherwise specified (see 6.2.1), First Article inspection shall comprise the examination in 4.4 and all tests in 4.5. Failure to pass the examination or any test shall be cause for disapproval of the First Article.

* 4.3 Quality conformance inspection. Quality conformance inspection shall be applied to each item prior to being offered for acceptance under the contract. Unless otherwise specified (see 6.2.1), quality conformance inspection shall consist of the examination in 4.4, the tests in 4.5 through 4,5.3 and the inspection in 4.6. Failure of the item to pass any examination or test shall be cause for rejection.

4.4 Examination. The machine and equipment shall be examined to determine compliance with the requirements in 3.2 through 3.5 and 3.7 through 3.16.

4.5 Tests.

4.5.1 Operational test. The machine shall be operated at no load for not less than 30 minutes. Proper operation of all controls, motors, adjusting mechanisms and accessories shall be verified during the trial period.

4.5.2 Honing accuracy test - Type I. The honing accuracy test shall be conducted on a gray iron casting not less than 4 inches square by 3 inches thick. A 1.000 inch diameter hole shall be bored and reamed through the 4 x 4 inch faces. The honing machine shall be used to finish the hole size to 1.005 inches, to include both roughing and finishing operations.

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The finish honing operation shall remove 0.0005 to 0.0010 inch of stock from the inside diameter. The roundness, straightness, and surface finish of the finished bore shall conform to the requirements of 3.6. Time required for actual honing shall not exceed ten minutes.

* 4.5.3 Honing accuracy test - Type II. Two workplaces shall be prepared of SAE 4130 steel, hardened to not less than Rockwell C-35, not less than 3 inches long by 1½ inch outside diameter, and with a 1.000 inch diameter hole bored and reamed through the axis of each. The first sample shall be honed to the dimensions and tolerances specified in 3.6. The honing speed, type of stones and type of mandrel used shall be at the option of the supplier. The experience gained in honing the first sample shall be used to assist in honing the second sample to finished size. The bore of the second sample shall be similarly honed to size. During the honing operation of the second sample, the bore may be measured not more than two times during the rough honing operation and not more than two times during the finish honing operation, using a precision gauging device other than the machine's sizing control. The sizing control dial may be readjusted after each measurement. After the second adjustment of the sizing dial during the finishing operation, the honing test shall be considered complete when the dial needle returns to zero. The dimensions and surface finish of the finished bore shall conform to the requirements of 3.6. Time required for actual honing of the second workpiece shall not exceed ten minutes for Style A and five minutes for Style B.

* 4,5.4 Electromagnetic interference control. Unless otherwise specified (see 6.2.1), equipment requiring electromagnetic interference control testing shall be tested for compliance with 3.1 in accordance with Table II of MIL-STD-461.

4.6 Inspection of preparation for delivery. Preservation-packaging, packing, and marking shall be inspected to determine compliance with the requirements of Section 5.

5. PREPARATION FOR DELIVERY

* 5.1 Preservation-packaging. Preservation-packaging shall be in accordance with MIL-M-18058, at Level A or C as specified (see 6.2.1).

5.2 Packing Packing shall be in accordance with MIL-M-18058, at Level A, B, or C as specified (see 6.2.1).

5.3 Marking. Marking shall comply with MIL-STD-129. Additional marking shall be as specified (see 6.2.1).

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6. NOTES

6.1 Intended use. Type I and Type II, Style A honing machines covered by this specification are intended for light, internal honing of metal parts, when stroking of the parts can be performed manually. Type II, Style B machines are intended for honing jobs where fast stock removal without physical effort is required.

6.2 Ordering data.

6.2.1 Procurement requirements. Purchasers should specify their requirements in procurement documents, including whether each choice is required or not required, by entering an appropriate statement identified to each of the following:

a. Title, number, and date of this specification.

b. Type, style and size machine required (see 1.2).

c. First Article approval, if required (see 3.1).

d. Specify exceptions and additional safety and health requirements (see 3.2.2 and 6.5).

e. Painting, if different, and Machine Tool Light Gray if required (see 3.3.5).

f. Specify hydraulic or mechanical stone expansion system (see 3.4.3).

g. Pump drive required; also machine characteristics, if different (see Table 1).

h. Specify if power filtering is required (see 3.4.6.1).

i. Electrical equipment, if different (see 3.7, 3.7.1 and 3.7.2).

j. Number of speeds and type of enclosure for motors, if different (see 3.7.3).

k. Elapsed time indicator, when required (see 3.7.4).

1. Standard equipment, if different (see 3.9).

m. Optional equipment, if required (see 3.10).

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n. Optional tooling required; specify quantity of mandrels together with sizes of bores to be honed and maximum honing length; specify quantity of stones together with type, dimensions, and composition; specify quantity of truing sleeves and diameters; fully describe other optional tooling required (see 3.10.1).

o. Electromagnetic interference control, if required (see 3.11).

p. Type fungus proofing, if required (see 3.12).

q. Lubrication chart, if different (see 3.13).

r. Nameplate, if different (see 3.14).

s. First Article inspection, if different (see 4.2).

t. Quality conformance inspection, if different (see 4.3).

u. Specify electromagnetic interference control test requirements, if different (see 4.5.4 and 6,4).

v. Required level of preservation-packaging (see 5.1).

w. Required level of packing (see 5.2).

x. Additional marking, if required (see 5.3).

* 6.2.2 Contract data requirements. Required technical data, such as operators manual, parts lists, foundation and anchor bolt plans, wiring diagrams and other instructions for operation and maintenance, as identified on a numbered DD Form 1664, should be specified on a DD Form 1423 incorporated into the contract.

* 6.3 First Article. When First Article approval is required under 3.1, the solicitation should adhere to the applicable portions of Section 1, Part 19 of the Armed Services Procurement Regulation.

* 6.4 Army and Air Force procurements. When equipment purchased for the Department of the Army or the Air Force requires electromagnetic interference control testing, such testing should be specified to be in accordance with MIL-STD-461, Table II, Class IIB equipment, except only Test Methods CE03 and RE02 should be required.

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* 6.5 Safety and health determinations. In order that equipment integrated into the user's operational environment will comply with the standards promulgated under the Occupational Safety and Health Act of 1970, limitation and control of noise levels, radiation, electromagnetic emission, noxious vapors, heat etc. as applicable, specific requirements concerning guarding the point(s) of operation, and other safety requirements should be specified.

* 6.6 Supersession data. This specification includes the requirements of and supersedes Military Specification MIL-H-14253C(WC) dated 11 February 1970. The designation is Type I, Size 312.

* 6.7 Size identification. The machines covered by this specification are sized by bore diameter x bore length.

* 6.8 Mercury prohibited. When the use of mercury or its compounds in the machine and its components and accessories is to be prohibited, procurement documents should include a statement to that effect.

6.9 Marginal notations. The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:
Army -WC
Navy - SH
Air Force - 84

Review Activities:
Army -WC
Navy - SH
DSA - GS

User Activities:
Army - MC
Navy - AS, MC, YD

Preparing Activity:
DSA-IP

Project Number:
3419-0160

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